Assignment 7.1

Week 7

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AIT620

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jupyter UN_Immigration_Analysis Last Checkpoint: 5 minutes ago

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Markdown JupyterLab Python 3 (ipykernel)

[1]:
```
import pandas as pd
```

[4]:
```
import os
print(os.getcwd())
print(os.listdir())
```

```
/Users/macminihique
['Untitled10.ipynb', 'Untitled7.ipynb', '.config', 'Music', 'Untitled12.ipynb', 'Untitled5.ipynb', 'paulo_abdala_dir.tar.gz', 'Assignment_
6_1 4.02.ipynb', 'Untitled1.ipynb', 'AIT620_Final_Wine_Quality2.pdf', '.DS_Store', '.CFUserTextEncoding', '.xonshrc', 'anaconda_projects',
'Untitled3.ipynb', 'Untitled.ipynb', '.zshrc', 'AIT620_Final_Wine_Quality2.ipynb', 'Untitled4.ipynb', 'Untitled6.ipynb', 'Pictures', 'Unti
tled11.ipynb', 'first_notebook.ipynb', '.zprofile', 'AIT620_template.ipynb', '.zsh_history', 'Untitled2.ipynb', '.ipython', 'Desktop', 'Li
brary', '.matplotlib', 'Activity_4_01_Car_Classification', '.cups', '.lacuna-webpki', 'Public', 'Assignment_6_1.ipynb', 'AIT620_Final_Exam
_Wine_Quality.ipynb', '.tcshrc', '.anaconda', 'Untitled8.ipynb', '.ssh', 'Movies', 'Applications', '.Trash', '.ipynb_checkpoints', '.jupyt
er', 'Documents', '.bash_profile', '.softplan-websigner', 'paulo', 'Downloads', '.continuum', 'paulo_abdala_dir', 'Untitled9.ipynb', '.pje
office-pro', '.zsh_sessions', '.conda']
```

[9]:
```
import os

print("Current folder:", os.getcwd())
print("Files:", os.listdir())
```

```
Current folder: /Users/macminihique
Files: ['Untitled10.ipynb', 'Untitled7.ipynb', '.config', 'Music', 'Untitled12.ipynb', 'Untitled5.ipynb', 'paulo_abdala_dir.tar.gz', 'Assi
gnment_6_1 4.02.ipynb', 'Untitled1.ipynb', 'AIT620_Final_Wine_Quality2.pdf', '.DS_Store', '.CFUserTextEncoding', '.xonshrc', 'anaconda_pro
jects', 'Untitled3.ipynb', 'Untitled.ipynb', 'Canada.xlsx', '.zshrc', 'AIT620_Final_Wine_Quality2.ipynb', 'Untitled4.ipynb', 'Untitled6.ip
ynb', 'Pictures', 'Untitled11.ipynb', 'first_notebook.ipynb', '.zprofile', 'AIT620_template.ipynb', '.zsh_history', 'Untitled2.ipynb', '.i
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nb', 'AIT620_Final_Exam_Wine_Quality.ipynb', '.tcshrc', '.anaconda', 'Untitled8.ipynb', '.ssh', 'Movies', 'Applications', '.Trash', '.ipyn
b_checkpoints', '.jupyter', 'Documents', '.bash_profile', '.softplan-websigner', 'paulo', 'Downloads', '.continuum', 'paulo_abdala_dir', '
Untitled9.ipynb', '.pjeoffice-pro', '.zsh_sessions', '.conda']
```

[10]:
```
import glob

files = glob.glob("/home/oai/share/**/canada*", recursive=True)
files += glob.glob("/home/jovyan/**/canada*", recursive=True)
files += glob.glob("/mnt/data/**/canada*", recursive=True)

print(files)
```

```
[]
```

[12]:
```
import pandas as pd
from pathlib import Path

downloads = Path("/Users/macminihique/Downloads")

files = list(downloads.glob("*anada*.xls*"))
print(files)
```

```
[PosixPath('/Users/macminihique/Downloads/Canada.xlsx')]
```

[13]:
```
df = pd.read_excel(
    files[0],
    sheet_name="Canada by Citizenship",
    skiprows=20
)

df.head()
```

[13]:

	Type	Coverage	OdName	AREA	AreaName	REG	RegName	DEV	DevName	1980	...	2004	2005	2006	2007	2008	2009	2010	2011	2
0	Immigrants	Foreigners	Afghanistan	935	Asia	5501	Southern Asia	902	Developing regions	16	...	2978	3436	3009	2652	2111	1746	1758	2203	2

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```
df.head()
```

[13]:

	Type	Coverage	OdName	AREA	AreaName	REG	RegName	DEV	DevName	1980	...	2004	2005	2006	2007	2008	2009	2010	2011	2
0	Immigrants	Foreigners	Afghanistan	935	Asia	5501	Southern Asia	902	Developing regions	16	...	2978	3436	3009	2652	2111	1746	1758	2203	2
1	Immigrants	Foreigners	Albania	908	Europe	925	Southern Europe	901	Developed regions	1	...	1450	1223	856	702	560	716	561	539	
2	Immigrants	Foreigners	Algeria	903	Africa	912	Northern Africa	902	Developing regions	80	...	3616	3626	4807	3623	4005	5393	4752	4325	
3	Immigrants	Foreigners	American Samoa	909	Oceania	957	Polynesia	902	Developing regions	0	...	0	0	1	0	0	0	0	0	
4	Immigrants	Foreigners	Andorra	908	Europe	925	Southern Europe	901	Developed regions	0	...	0	0	1	1	0	0	0	0	

5 rows × 43 columns

[14]:

```
# Rename columns
df.rename(columns={
    "OdName": "Country",
    "AreaName": "Continent",
    "RegName": "Region"
}, inplace=True)

# Select required columns
years = list(range(1980, 2014))
columns = ["Country", "Continent", "Region", "DevName"] + years

df = df[columns].copy()

# Calculate total immigration
df[years] = df[years].apply(
    pd.to_numeric, errors="coerce"
).fillna(0).astype(int)

df["Total"] = df[years].sum(axis=1)

# Set Country as index
df.set_index("Country", inplace=True)
df.index.name = "Country"

# Display final table
display(df.head())
```

Country	Continent	Region	DevName	1980	1981	1982	1983	1984	1985	1986	...	2005	2006	2007	2008	2009	2010	2011	2012	2013
Afghanistan	Asia	Southern Asia	Developing regions	16	39	39	47	71	340	496	...	3436	3009	2652	2111	1746	1758	2203	2635	2004
Albania	Europe	Southern Europe	Developed regions	1	0	0	0	0	0	1	...	1223	856	702	560	716	561	539	620	603
Algeria	Africa	Northern Africa	Developing regions	80	67	71	69	63	44	69	...	3626	4807	3623	4005	5393	4752	4325	3774	4331
American Samoa	Oceania	Polynesia	Developing regions	0	1	0	0	0	0	0	...	0	1	0	0	0	0	0	0	0
Andorra	Europe	Southern	Developed	0	0	0	0	0	0	2	...	0	1	1	0	0	0	0	1	1

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```
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Country	Continent	Region	DevName	1980	1981	1982	1983	1984	1985	1986	...	2005	2006	2007	2008	2009	2010	2011	2012	2013
Afghanistan	Asia	Southern Asia	Developing regions	16	39	39	47	71	340	496	...	3436	3009	2652	2111	1746	1758	2203	2635	2004
Albania	Europe	Southern Europe	Developed regions	1	0	0	0	0	0	1	...	1223	856	702	560	716	561	539	620	603
Algeria	Africa	Northern Africa	Developing regions	80	67	71	69	63	44	69	...	3626	4807	3623	4005	5393	4752	4325	3774	4331
American Samoa	Oceania	Polynesia	Developing regions	0	1	0	0	0	0	0	...	0	1	0	0	0	0	0	0	0
Andorra	Europe	Southern Europe	Developed regions	0	0	0	0	0	0	2	...	0	1	1	0	0	0	0	1	1

5 rows × 38 columns

UN Immigration Data Transformation

This project uses United Nations immigration data for Canada from 1980 to 2013.

Using Python and pandas, I cleaned and transformed the dataset by renaming columns, selecting relevant information, setting countries as the index, and calculating total immigration for each country.

The final DataFrame matches the required format.